Fractal Dimension: Theory and Computation

Box-Counting Estimation via Hilbert Curve Analysis

ME

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Contents

1 Introduction

This entire paper is basically based on [Fal03]. This is just what I found when I wanted to look into this area.

Classical dimension is usually viewed at as an integer e.g. a square is 2D. However, is there a way to consider dimension in another manner? Geometric objects in nature (e.g. coastlines, trees, etc.) exhibit self similarity at all scales and resist this classification. How do we measure the "size" for such a set? Is there a scaling property associated with dimension?

This short write-up develops the measure theory needed to understand said property (Hausdorff measure and dimension), and introduces ways to compute measure (through the box-counting dimension). We use the Hilbert curve as a case study in which we can analytically derive the dimension and computationally estimate it.

2 Measure Theory

The material in this section is standard and follows the presentation in [Men26].

First to define measure theory (I didn't take Lebesgue so I will be using the above notes...)

A measure is a way to assign a "size" to subsets of a set X. Let $\mathcal{F}$ be a σ-algebra on X.

Definition 2.1. (Measure). Let $(X, \mathcal{F})$ be a measurable space. A measure on (X, $\mathcal{F}$) is a function $\mu : \mathcal{F} \to [0, \infty]$ satisfying:

1. $\mu(\emptyset) = 0$
2. **Countable additivity:** if $A_1, A_2, \ldots \in \mathcal{F}$ are pairwise disjoint, then
$$\mu\left(\bigsqcup_{i=1}^{\infty} A_i\right) = \sum_{i=1}^{\infty} \mu(A_i)$$

The triple $(X, \mathcal{F}, \mu)$ is called a **measure space**.

Proposition 2.2 (Basic properties)**.** *Let $(X, \mathcal{F}, \mu)$ be a measure space.*

1. ***Monotonicity**:*
$$A \subseteq B \Rightarrow \mu(A) \leq \mu(B)$$
2. ***Countable subadditivity**:*
$$\mu\left(\bigcup_{i=1}^{\infty} A_i\right) \leq \sum_{i=1}^{\infty} \mu(A_i)$$
3. ***Continuity from below**: if $A_1 \subseteq A_2 \subseteq \cdots$ then*
$$\mu\left(\bigcup_{i=1}^{\infty} A_i\right) = \lim_{i\to\infty} \mu(A_i)$$
4. ***Continuity from above**: if $A_1 \supseteq A_2 \supseteq \cdots$ and $\mu(A_1) < \infty$ then*
$$\mu\left(\bigcap_{i=1}^{\infty} A_i\right) = \lim_{i\to\infty} \mu(A_i)$$

The canonical example is the **Lebesgue measure** λ^n on $(\mathbb{R}^n, \mathcal{B}(\mathbb{R}^n))$, where $\mathcal{B}(\mathbb{R}^n)$ is the Borel σ-algebra — the smallest σ-algebra containing all open sets. Lebesgue measure assigns to each box $[a_1, b_1] \times \cdots \times [a_n, b_n]$ its volume $(b_1 - a_1) \cdots (b_n - a_n)$, and extends uniquely to all Borel sets. It formalises the intuitive notions of length, area, and volume.

The inadequacy of Lebesgue measure for fractal geometry is immediate: a curve $\gamma \subset \mathbb{R}^2$ has $\lambda^2(\gamma) = 0$ (zero area) but $\lambda^1(\gamma)$ may be infinite (infinite length). Neither gives useful information about the "size" of γ. What is needed is a family of measures parametrised by a continuous exponent $s \geq 0$.

To construct such a family, we need the notion of an outer measure, which is slightly weaker than a measure — it is defined on *all* subsets of X, not just measurable ones, and only requires subadditivity rather than full additivity.

Definition 2.3 (Outer measure)**.** A function $\mu^* : 2^X \to [0, \infty]$ is an **outer measure** on X if:

1. $\mu^*(\emptyset) = 0$
2. **Monotonicity**: $A \subseteq B \Rightarrow \mu^*(A) \leq \mu^*(B)$
3. **Countable subadditivity**:

$$\mu^*\left(\bigcup_{i=1}^{\infty} A_i\right) \leq \sum_{i=1}^{\infty} \mu^*(A_i)$$

The relationship between outer measures and measures is given by the Hahn-Carathéodory extension theorem: given an outer measure μ^* on X, the collection

$$\Sigma = \{A \subseteq X : \mu^*(B) = \mu^*(B \cap A) + \mu^*(B \setminus A) \text{ for all } B \subseteq X\}$$

is a σ-algebra, and $\mu^*|_\Sigma$ is a measure. Sets in Σ are called **μ^*-measurable**. In particular, all Borel sets are μ^*-measurable whenever μ^* is constructed from a Borel-compatible pre-measure — which will be the case for Hausdorff measure.

The standard way to construct an outer measure is via coverings. If $\mathcal{K} \subseteq 2^X$ covers X (with $\emptyset \in \mathcal{K}$) and $\tilde{\mu} : \mathcal{K} \to [0, \infty]$ assigns sizes to sets in $\mathcal{K}$ with $\tilde{\mu}(\emptyset) = 0$, then

$$\mu^*(A) = \inf\left\{\sum_{i=1}^{\infty} \tilde{\mu}(K_i) \;\middle|\; K_i \in \mathcal{K},\ A \subseteq \bigcup_{i=1}^{\infty} K_i\right\}$$

defines an outer measure on X. This is precisely the construction used to define Hausdorff measure — with $\mathcal{K}$ being all subsets of diameter at most δ, and $\tilde{\mu}(U) = |U|^s$.

3 Hausdorff Measure

The material in sections 3 and 4 follows [Fal03].

3.1 Definition

Let (X, d) denote a metric space, $S \subseteq X$ a Borel set.
For a set $U \subseteq X$, write $|U| := \operatorname{diam}(U) = sup_{x,y \in U} d(x, y)$

Definition 3.1. Let $s \geq 0$, and $\delta > 0$. Define

$$\mathcal{H}^s_\delta(S) = \inf \left\{ \sum_{i=1}^{\infty} |U_i|^s \;\middle|\; S \subseteq \bigcup_{i=1}^{\infty} U_i, \;\; |U_i| \leq \delta \right\}$$

This is the infimum over all countable covers of S by sets of diameter at most δ.

As $\delta \to 0$, we allow fewer and fewer covers, so $\mathcal{H}^s_\delta(S)$ is non-decreasing in $1/\delta$. The limit

$$\mathcal{H}^s(S) = \lim_{\delta \to 0} \mathcal{H}^s_\delta(S) \in [0, \infty]$$

is the s-dimensional Hausdorff measure of S. This is possibly $+\infty$.
$\mathcal{H}^s$ is an outer measure on X, and its restriction to the Borel σ-algebra is a measure.

3.2 Phase Transition

Observe that $\mathcal{H}^s(S)$ exhibits a phase transition in s:

Theorem 3.2. *If $\mathcal{H}^s(S) < \infty$ for some s, then $\mathcal{H}^t(S) = 0$ for all $t > s$.*

Proof. Suppose $\mathcal{H}^s(S) < \infty$. For any cover $\{U_i\}$ with $|U_i| \leq \delta$. For $t > s$:

$$\Sigma |U_i|^t = \Sigma |U_i|^{t-s} \cdot |U_i|^s \leq \delta^{t-s} \Sigma |U_i|^s$$

Taking the infimum over all covers gives

$$\mathcal{H}^t_\delta(S) \leq \delta^{t-s} \cdot \mathcal{H}^s_\delta(S) \leq \delta^{t-s} \cdot \mathcal{H}^s(S)$$

where the second inequality holds since $\mathcal{H}^s_\delta(S)$ is non-decreasing in $1/\delta$.
Now let $\delta \to 0$, $t - s > 0$, and $\mathcal{H}^s(S) < \infty$ is a fixed finite constant. So the right hand side goes to 0, hence $\mathcal{H}^t(S) = 0$. $\square$

Corollary 3.3. *There exists a unique value $d \in [0, \infty]$ such that*

$$\mathcal{H}^s(S) = \begin{cases} \infty & s < d \\ 0 & s > d \end{cases}$$

At $s = d$, the measure $\mathcal{H}^d(S)$ may take any value in $[0, +\infty]$.

4 Hausdorff Dimension

Definition 4.1. (Hausdorff Dimension). The **Hausdorff Dimension** of S is the critical exponent

$$\dim_{\mathcal{H}}(S) = \inf\{s \geq 0 : \mathcal{H}^s(S) = 0\} = \sup\{s \geq 0 : \mathcal{H}^s(S) = +\infty\}$$

This is the "true" dimension of a fractal.

4.1 Basic Properties

Proposition 4.2 (Basic properties)**.** *Hausdorff dimension satisfies the following:*

1. ***Monotonicity**: $A \subseteq B \implies \dim_{\mathcal{H}}(A) \leq \dim_{\mathcal{H}}(B)$*
2. ***Countable stability**: $\dim_{\mathcal{H}}(\cup_{i=1}^{\infty} A_i) = \sup_i \dim_{\mathcal{H}}(A_i)$*
3. ***Countable sets**: if S is countable, $\dim_{\mathcal{H}}(S) = 0$.*
4. ***Smooth manifolds** [1]: if $M \subset \mathbb{R}^n$ is a smooth k-dimensional submanifold, then $\dim_{\mathcal{H}}(M) = k$.*

The following proposition is a useful tool we'll need later down the line:

Proposition 4.3. *(Lipschitz invariance). Let $f : A \to B$ be Lipschitz with constant C. Then $\dim_{\mathcal{H}}(f(A)) \leq \dim_{\mathcal{H}}(A)$. In particular, bi-Lipschitz maps preserve Hausdorff dimensions.*

Proof. If $\{U_i\}$ covers A with $\Sigma|U_i|^s < \infty$, then $\{f(U_i)\}$ covers $f(A)$ with $|f(U_i)| \leq C|U_i|$ (by Lipschitz), so $\Sigma|f(U_i)|^s \leq C^s\Sigma|U_i|^s < \infty$ Hence $\mathcal{H}^s(f(A)) \leq C^s\mathcal{H}^s(A)$, and the dimension inequality follows. $\square$

[1] A smooth k-dimensional manifold is just a set that locally looks like $\mathbb{R}^k$

5 Box-counting Dimension

5.1 Definition

A.K.A Minkowski–Bouligand dimension.
We need a way to be able to effectively compute the dimension of a fractal (i.e. the Hausdorff dimension). A perhaps easy way to do this is to cover said shape with boxes of size ε and count them. We check to see how the number of boxes hitting the shape $N(\varepsilon)$ changes w.r.t the length ε. We use the following formula:

$$\dim_{\text{box}}(S) := \lim_{\varepsilon \to 0} \frac{\log N(\varepsilon)}{\log(1/\varepsilon)} = -\lim_{\varepsilon \to 0} \frac{\log N(\varepsilon)}{\log(\varepsilon)}$$

This is roughly the dimension d such that $N(\varepsilon) \approx C\varepsilon^{-d}$.
Intuitively, we choose this to represent the dimension because a d-dimensional box of side ε has a "d-dimensional volume" ε^d. So we end up with some $N(\varepsilon) \approx \frac{V}{\varepsilon^d} = C\varepsilon^{-d}$. We want to find a d for a fractal i.e. finding the d such that the same scaling property holds.

5.2 Relationship to Hausdorff dimension

Proposition 5.1. *For any bounded $S \subseteq \mathbb{R}^n$, $\dim_{\mathcal{H}}(S) \leq \dim_{\text{box}}(S)$.*

Proof. Let $d > \dim_{\text{box}}(S)$. Then $N(\varepsilon) \leq C\varepsilon^{-\dim_{\text{box}}(S)}$ for some constant C and all small ε. Covering S with $N(\varepsilon)$ hypercubes of diameter $\varepsilon\sqrt{n}$:

$$\mathcal{H}^d_{\varepsilon\sqrt{n}}(S) \leq N(\varepsilon) \cdot (\varepsilon\sqrt{n})^d \leq C(\sqrt{n})^d \cdot \varepsilon^{d - \dim_{\text{box}}(S)} \to 0$$

as $\varepsilon \to 0$. Hence $\mathcal{H}^d(S) = 0$, so $\dim_{\mathcal{H}}(S) \leq d$. Since $d > \dim_{\text{box}}(S)$ was arbtirary, $\dim_{\mathcal{H}}(S) \leq \dim_{\text{box}}(S)$. $\square$

This is a very useful property as will be shown in the next few chapters. With this we now have a best upper bound for calculting the Hausdorff dimension. The equality actually holds on "well-behaved" fractals which is exactly what we need.

6 Self-similar sets and Moran's theorem

This section explores [FGS16].

6.1 Iterated Function Systems

Definition 6.1. (IFS and attractor). An **iterated function system** (IFS) is a finite collection of contractions[2] $\{f_1, ..., f_N\}$ on a complete[3] metric space (X, d), where each f_i has Lipschitz constant $r_i \in (0,1)$. The map $F(K) = \cup_{i=1}^{N} f_i(K)$ on the space of compact[4] subsets of X (using the Hausdorff metric - $d_H(A, B) = \max(\sup_{a\in A} d(a, B), \sup_{b\in B} d(b, A)))$ is itself a contraction, so by the Banach fixed-point theorem there exists a unique compact **attractor** $K \subseteq X$ such that:

$$K = \bigcup_{i=1}^{N} f_i(K)$$

Definition 6.2. (Open set condition). The IFS $f_1, ..., f_N$ satisfies the **open set condition** (OSC) if there exists a non-empty bounded open set $V \subseteq X$ such that:

$$\bigcup_{i=1}^{N} f_i(V) \subseteq V, \quad f_i(V) \cap f_j(V) = \emptyset \text{ for } i \neq j$$

This ensures the pieces of K do not overlap in a measure-theoretically significant way, which is necessary for the dimension formula to hold. Basically, any overlap would cause the dimension estimate to corrupt.

6.2 Moran's theorem

Theorem 6.3. *(Moran, 1946). Let $\{f_1, ...f_N\}$ be an IFS of similarities on $\mathbb{R}^n$ with contraction ratios $r_1, ...r_N \in (0,1)$, satisfying the OSC. Let K be its attractor. Then*

$$\dim_{\mathcal{H}}(K) = \dim_{\text{box}}(K) = s,$$

where s is the unique solution to the ***Moran equation***

$$\sum_{i=1}^{N} r_i^s = 1.$$

Uniqueness of s follows from the fact that $\varphi(s) = \Sigma_i r_i^s$ is strictly decreasing in s, with $\varphi(0) = N > 1$ and $\varphi(\infty) = 0$.
For a uniform IFS ($\forall i,\ r_i = r \in (0,1)$), the Moran equation gives $Nr^s = 1$, hence $s = \log N / \log(1/r)$, which is the similarity dimension. A uniform IFS applies to all of your nice looking fractals. A non-uniform IFS (e.g. coastline) would have to have a numerical method of solving for this such as bisection or Newton's method.

[2] A contraction f is a function such that $d(f(x), f(y)) \leq \alpha \cdot d(x, y)$, where $\alpha \in [0,1)$.
[3] A complete metric space is one such that every Cauchy sequence converges.
[4] Compact sets are closed and bounded.

6.3 Examples

Example 6.4. (Cantor set). There are two similarities on $[0, 1]$, each with ratio $r = 1/3$. We know this since at each iteration each line segment is replaced by 2 line segments, each a third of the length of the original. By Moran's theorem: $2 \cdot (1/3)^s = 1$, so $s = \log 2 / \log 3 \approx 0.63092975$

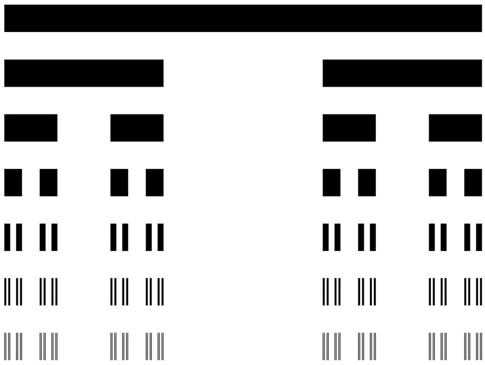


Figure 1: Cantor set construction (takes all values that aren't removed). [The17]

Example 6.5. (Serpiński triangle). There are three similarities each with ratio $r = 1/2$. By Moran's theorem: $3 \cdot (1/2)^s = 1$, so $s = \log 3 / \log 2 \approx 1.58496250$

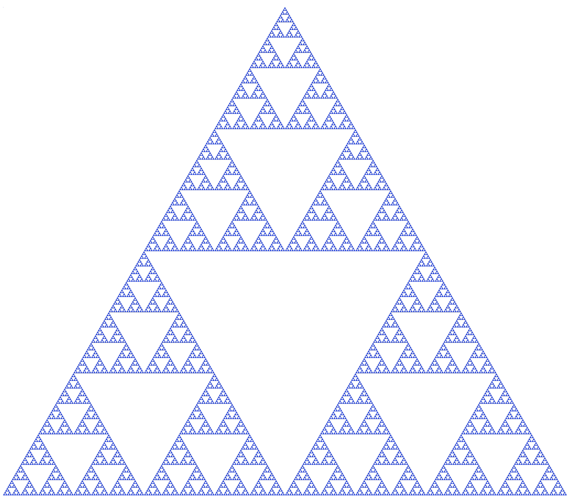


Figure 2: Serpiński triangle. [Wik]

Example 6.6. (Koch curve). There are four similarities each with ratio $r = 1/3$. By Moran's theorem: $4 \cdot (1/3)^s = 1$, so $s = \log 4 / \log 3 \approx 1.26185951$

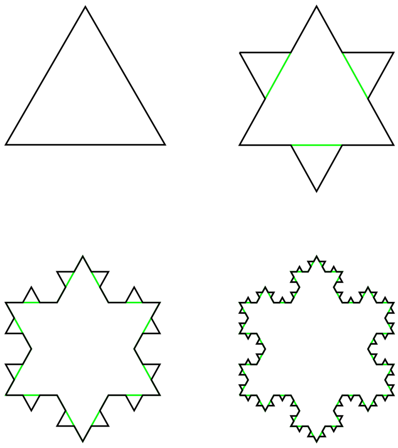


Figure 3: Koch snowflake with 4 iterations. [CSW]

7 L-systems

Moran's theorem gives the dimension of an IFS attractor, but specifying an IFS by hand is long. L-systems[5] provide a purely symbolic description of the same objects, from which λ and k can be recovered almost mechanically. This gives us a family of fractals with exactly known dimension, which we will use in §9 to validate the box-counting estimator.

7.1 Definition

Definition 7.1. (D0L-system). A **deterministic context-free L-system** is a triple $G = (V, \omega, P)$ where V is a finite alphabet, $\omega \in V^+$ is the **axiom**, and $P : V \to V^*$ assigns a **production** to each symbol. Writing $\omega_0 = \omega$, the word ω_{n+1} is obtained from ω_n by replacing *every* symbol simultaneously by its production.

We need parallel writing here so that every part of the word changes at the same rate for self-similarity.

7.2 Turtle interpretation

Definition 7.2. (Turtle interpretation). Fix a step length $\ell > 0$ and a turning angle δ. A turtle has state $(x, y, \alpha) \in \mathbb{R}^2 \times [0, 2\pi)$. Reading a word left to right:

- `F`: move forward ℓ, drawing a segment;
- `f`: move forward ℓ without drawing;
- `+`, `-`: turn left, right by δ.

Symbols with no geometric interpretation are ignored. Write γ_n for the drawn figure of ω_n with step length ℓ_n.

Note that the grammar alone does not determine a curve: the angle δ and the choice of ℓ_n are geometric data supplied separately.

7.3 The growth matrix

Definition 7.3. (Substitution matrix). For $G = (V, \omega, P)$ with $V = \{a_1, \ldots, a_m\}$, the **substitution matrix** $M \in \mathbb{N}^{m \times m}$ is given by

$$M_{ij} = \#\{\text{occurrences of } a_j \text{ in } P(a_i)\}.$$

7.4 Growth of the substitution matrix

Let $v_n \in \mathbb{N}^m$ record the symbol counts of ω_n, so that $v_{n+1} = v_n M$ and hence $v_n = v_0 M^n$. Since box-counting sees only the drawn segments, the quantity of interest is the `F`-component of v_n.

[5] Introduced by Lindenmayer in 1968 tomodel plant growth; see [PL90].

Theorem 7.4. *(Perron–Frobenius). Let M be a square matrix with non-negative entries and spectral radius $\lambda = \max_i |\lambda_i|$. Then λ is an eigenvalue of M, with a non-negative eigenvector. If moreover M is irreducible, then λ is simple, satisfies $\lambda > |\lambda_i|$ for every other eigenvalue, and its eigenvector may be taken strictly positive.*

Expanding v_0 in a basis of left eigenvectors gives $v_n = \sum_i c_i \lambda_i^n u_i$; dividing by λ^n shows every subdominant term vanishes, so $v_n \sim c_1 \lambda^n u_1$ and $\|v_n\| = \Theta(\lambda^n)$. Perron–Frobenius is what rules out a complex or sign-alternating dominant eigenvalue, which would make $\|v_n\|$ oscillate rather than converge.

Example 7.5. (Hilbert). The Hilbert curve is generated by

$$A \to \texttt{+BF-AFA-FB+}, \qquad B \to \texttt{-AF+BFB+FA-}, \qquad \delta = 90^\circ,$$

with $\texttt{F} \to \texttt{F}$. Here A and B carry the recursion and are ignored by the turtle; two are needed because the four sub-cells occur in both handednesses. On (A, B, F),

$$M = \begin{pmatrix} 2 & 2 & 3 \\ 2 & 2 & 3 \\ 0 & 0 & 1 \end{pmatrix}.$$

With $v_0 = (1, 0, 0)$ the counts are $v_1 = (2, 2, 3)$, $v_2 = (8, 8, 15)$, $v_3 = (32, 32, 63)$, so the number of drawn segments at level n is $4^n - 1$. The eigenvalues of M are $4, 1, 0$, and correspondingly $F_n = 1 \cdot 4^n - 1 \cdot 1^n$: the dominant eigenvalue sets the growth rate and the subdominant one contributes the correction. Note $v_3 \approx 32 \cdot (1, 1, 2)$, the dominant left eigenvector.

Remark 7.6. (Reducibility). The matrix above is reducible: the associated digraph has edges $A, B \to F$ but none from F back, so it is not strongly connected and the strong form of 7.4 does not apply. The conclusion survives because `F` is reachable from the block $\{A, B\}$ carrying the dominant eigenvalue. In general it suffices that the drawn symbols be reachable from the dominant block.

Proposition 7.7. *(Dimension of a D0L-system). Let G be a D0L-system with substitution matrix M, let λ be the spectral radius of M, and suppose the turtle interpretation admits a scaling factor $k > 1$, meaning that γ_n is drawn with step length $\ell_n = k^{-n}$ and the level-n figure is a union of $\Theta(\lambda^n)$ segments of that length. Then*

$$\dim_{\text{box}}(K) \le \frac{\log \lambda}{\log k},$$

where $K = \lim_n \gamma_n$, with equality when the induced IFS satisfies the OSC.

Proof. By Perron–Frobenius the number of drawn symbols in ω_n grows as $\Theta(\lambda^n)$. Covering

K at scale $\varepsilon = k^{-n}$ requires $N(\varepsilon) = O(\lambda^n)$ boxes, so

$$\frac{\log N(\varepsilon)}{\log(1/\varepsilon)} \leq \frac{n \log \lambda + O(1)}{n \log k} \longrightarrow \frac{\log \lambda}{\log k}.$$

Under the OSC the drawn segments overlap only at endpoints, so the count is $\Theta(\lambda^n)$ from below as well and the limit is attained. □

The content of 7.7 is a separation of concerns: λ is **combinatorial**, obtained from the grammar alone, whereas k is **geometric**, determined by δ and the net displacement of the turtle after one production. Neither determines the dimension by itself.

Remark 7.8. (Why the spectral radius, not a symbol count). For an edge-rewriting system such as Koch (`F` $\to$ `F+F-F+F`) one has $M = (4)$ and $\lambda = 4$, recovering the naive count of `F` symbols. For a *node*-rewriting system it fails: in Example 7.5 each production contains three `F`s, which would suggest $\log 3/\log 2 \approx 1.585$, whereas the spectral radius gives the correct $\log 4/\log 2 = 2$. The discrepancy is that the A and B symbols in each expansion themselves generate further `F`s at the next level.

7.5 A validation family

Fractal	Production(s)	δ	λ/k	Dimension
Cantor	`F`→`FfF`, `f`→`ff`	—	2/3	0.630930
Koch	`F`→`F+F-F+F`	60°	4/3	1.261860
Minkowski	`F`→`F+F-F-FF+F+F-F`	90°	8/4	1.5
Arrowhead	`A`→`B-A-B`, `B`→`A+B+A`	60°	3/2	1.584963
Peano	`F`→`F+F-F-F-F+F+F+F-F`	90°	9/3	2
Hilbert	see 7.5	90°	4/2	2

Table 1: L-systems with exactly known dimension, used in §9 as ground truth.

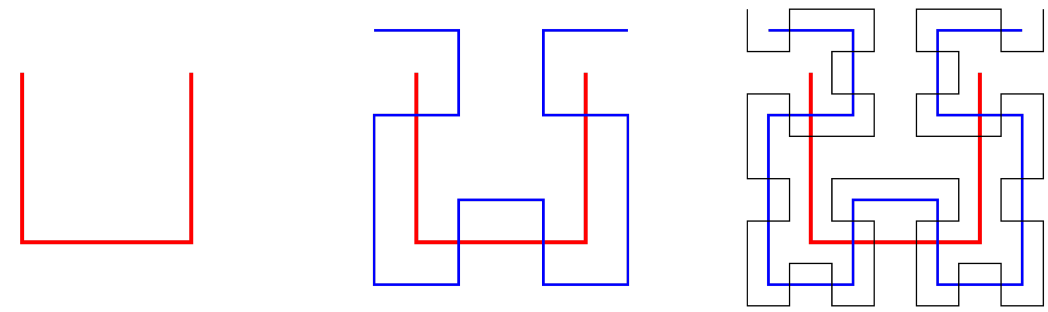
These span [0.63, 2] and are generated by a single implementation, so they provide a controlled test of the box-counting estimator against known values before it is applied to cases where the answer is not known in advance.

8 Hilbert curve

This is possibly my favourite fractal that I'd like to explore. It is essentially a mapping from 1D to 2D space (earning the title space-filling curve).

8.1 Construction

The Hilbert curve is a continuous surjection $\gamma : [0,1] \to [0,1]^2$ constructed as the uniform limit of a sequence of piecewise-linear approximations γ_n. At order n, the curve γ_n traverses all 4^n cells of a $2^n \times 2^n$ grid, connecting cell centres via a path that divides each grid cell into four sub-cells connected in a U-shape, rotated and reflected to ensure the overall path shape is continuous. From this we can see directly that it has dimension 2 (even though its not closed), since we have 4 similarities with ratio $1/2$, giving $\log 4 / \log 2 = 2$.



(a) First iteration. **(b)** Iterations 1 and 2. **(c)** Iterations 1 to 3.

Figure 4: Hilbert curve approximations. [Qef]

8.2 Properties of the limit curve

9 Computational Estimation

TODO. (code first, write later).

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